Python Computational Physics Problem Solving

Written Paper

HCMUT

General Physics 1 – CC01

Conservation of Linear Momentum in Collisions

Space Debris Impact on Solar Panels

Nguyễn Khánh Hưng – 2352435 (Group Leader)
Tăng Nguyễn Minh Duy – 2651233
Diệp Phú Thành – 2653138
Nguyễn Lâm Chí Tường – 2651313
Nguyễn Lê Anh Tuấn – 2651311
Châu Kiến Toàn – 2353192
Nguyễn Duy Thành – 2353101
Phạm Anh Minh – 2651271
Vũ Hồng Quân – 2453083

Instructor: Dr. Gary

November 6, 2026

ABSTRACT

Space debris is a growing danger for satellites, and solar panels are the parts that get hit most often because they are so large. In this project, *Space Debris Impact on Solar Panels*, we use conservation of linear momentum, $\mathbf{P}_{\text{before}} = \mathbf{P}_{\text{after}}$, to see how a small debris fragment changes a satellite's motion when it hits a panel. We want to find the impulse the satellite gets, its change in velocity, the spin it starts to have, and how much energy is lost, for different impact conditions. Our hypothesis is that the velocity change is tiny compared with the orbital speed, but the spin and the energy loss are not. A Python program uses the impulse–momentum theorem and conservation of linear and angular momentum to solve three scenarios, and it also runs sweeps over debris mass, impact speed, impact angle and mass ratio. We model the hit in two dimensions as an instant collision between a point-mass fragment and a rigid satellite. Two ideal cases are solved: an elastic collision with no friction, where the fragment bounces off, and a completely inelastic collision, where the fragment sticks to the panel. The spin comes from conservation of angular momentum about the centre of mass, and any kinetic energy that is not left as motion counts as lost. In the baseline case, a 1 g fragment at $10\,\mathrm{km\,s^{-1}}$ and 30° sticks to the panel 4 m from the centre of mass of a 500 kg satellite. The satellite's velocity changes by only $20.0\,\mathrm{mm\,s^{-1}}$ (2.6×10^{-6} of its orbital speed), but it starts spinning at $1.65^\circ\,\mathrm{s^{-1}}$, and 99.999% of the 50 kJ impact energy is lost as heat and damage. Momentum, angular momentum and energy balance to about 10^{-16}. The sweeps show that the velocity change grows with debris mass and speed. For a bouncing fragment it gets smaller as the hit becomes more grazing, while a sticking fragment gives the same velocity change at every angle. Our model only covers the momentum exchange right at the hit; holes, cracks and other damage to the panel are not included.

1. INTRODUCTION

The general topic of our project is conservation of linear momentum in collisions. When two objects hit each other, the forces between them are internal forces of the two-object system. During the very short contact they are also much bigger than any outside force, so the total momentum stays the same before and after the collision:

$$m\mathbf{v}_{d0} + M\mathbf{V}_0 = m\mathbf{v}_{d1} + M\mathbf{V}_1, \tag{1.1}$$

where m and M are the masses of the debris fragment and the satellite, $\mathbf{v}_d$ and $\mathbf{V}$ are their velocities, and the subscripts 0 and 1 mean before and after the hit. Equation (1.1) is true for every kind of collision. What changes from one kind to another is the kinetic energy: it is kept in an elastic collision and partly lost in an inelastic one (Halliday et al., 2014; Ling et al., 2016).

Our specific title is *Space Debris Impact on Solar Panels*. Low Earth orbit is full of small fragments left over from old explosions and collisions, and every new collision makes more of them. This chain reaction was first described by Kessler & Cour-Palais (1978). Solar panels are usually the biggest surfaces on a satellite, so they get hit the most. A real example happened on 23 August 2016, when a particle about 1 cm in size hit a solar panel of the Sentinel-1A satellite. The satellite lost part of its power, and its orbit and orientation changed suddenly (Krag et al., 2017). This event shows the two effects we study: a change in the satellite's velocity and a change in its spin.

Both effects can be explained with basic physics. The fragment gives part of its momentum to the satellite (Eq. (1.1)). Because the panel is far from the satellite's centre of mass, the same push also creates a torque, so the satellite starts to rotate. This is described by conservation of angular momentum. Kinetic energy is different, because it grows with v^2. A fragment moving at $10\,\mathrm{km\,s^{-1}}$ carries a lot of energy, and in a completely inelastic hit most of it turns into heat and damage on the panel.

To solve the problem, we model the satellite as a rigid body and the fragment as a point mass. We then solve the two limiting collisions from the course exactly. In the elastic case kinetic energy is kept, and in the completely inelastic case the fragment sticks to the panel and both move as one rigid body. Our Python program calculates the satellite's velocity change, its spin and the energy lost for three scenarios and several parameter sweeps, and compares the velocity change with the orbital speed. We check the code in two ways. First, the momentum, angular momentum and energy before and after the hit must be equal up to round-off error. Second, a head-on hit through the centre of mass must give the textbook 1-D results: with equal masses, the target should move at v_0 in the elastic case and at $v_0/2$ in the completely inelastic case (Ling et al., 2016). At the end we compare our results with what was measured for Sentinel-1A (Krag et al., 2017).

2. HYPOTHESIS

From conservation of linear and angular momentum, we expect that a fragment of about one gram moving at about $10\,\mathrm{km\,s^{-1}}$ will change the velocity of a 500 kg satellite by only about $mv_0/M \approx 20\,\mathrm{mm\,s^{-1}}$, which is very small (around 10^{-6}) compared with the orbital speed. However, if it hits the panel far from the centre of mass, the satellite should start to spin at a rate we can notice, around $1^\circ\,\mathrm{s^{-1}}$. We also expect that in a completely inelastic hit almost all of the fragment's kinetic energy (a fraction close to $M/(m+M)$) is lost as heat and damage.

3. BASIC PHYSICS

3.1. PHYSICAL SYSTEM AND REFERENCE FRAME

We describe the motion in a 2-D frame (Fig. 3.1). The x axis points away from Earth, y points along the direction of flight, and z (out of the plane) is perpendicular to the orbit, representing the spin axis.

The satellite (body and panels) is a rigid body with mass M (kg) and moment of inertia I (kg m^2) about the z axis through its centre of mass (CoM). In a frame moving with the satellite, it is at rest before the hit.

The fragment is a point mass m (kg) moving at speed v_0 (m s^{-1}). It hits the panel at the point $\mathbf{r}$ (m) (measured from the CoM), at an angle θ from the outward normal $\hat{\mathbf{n}}$ of the panel. With the panel tangent $\hat{\mathbf{t}} = \hat{\mathbf{z}} \times \hat{\mathbf{n}}$, the velocity of the fragment and its speed along the normal are:

$$\mathbf{v}_{d0} = -v_0 \left(\cos\theta\, \hat{\mathbf{n}} + \sin\theta\, \hat{\mathbf{t}}\right), \qquad v_n = v_0 \cos\theta. \tag{3.1}$$

So $\theta = 0$ is a head-on hit and θ close to 90° is a grazing hit.

- $\mathbf{v}_{d0}$: the initial velocity of the fragment (m s^{-1})
- v_n: the speed of the fragment along the normal (m s^{-1})

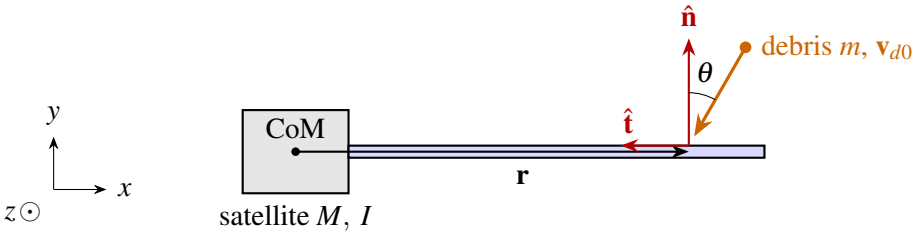


Figure 3.1: Geometry of the impact: a fragment of mass m hits the panel at lever arm $\mathbf{r}$ from the satellite's centre of mass, at angle θ from the panel normal $\hat{\mathbf{n}}$.

When the fragment hits the panel, it transfers momentum to the satellite. This can change both the satellite's linear and rotational motion. If the impact happens away from the centre of mass, it can also create a torque, causing the satellite to rotate. The impact angle affects how much of the fragment's velocity acts in the normal and tangential directions.

3.2. CONSERVATION OF LINEAR AND ANGULAR MOMENTUM

The contact lasts only microseconds, so the impulse from gravity ($\sim Mg\,\Delta t$) is tiny and we can treat the fragment and the satellite as an isolated system. This means its linear momentum and its angular momentum about the satellite's CoM do not change:

$$\mathbf{P} = m\mathbf{v}_d + M\mathbf{V} = \text{const}, \qquad L_z = m\,(\mathbf{r} \times \mathbf{v}_d)_z + I\omega = \text{const}, \tag{3.2}$$

where ω is the angular velocity of the satellite about z. So the momentum the fragment loses goes to the satellite, and since the fragment does not aim at the CoM, its angular momentum $m\mathbf{r}\times\mathbf{v}_d$ becomes the satellite's spin.

3.3. ELASTIC COLLISION: IMPULSE AND TORQUE

For the elastic collision, the friction is neglected, so the panel can act on the fragment only along the normal, delivering an impulse $J\hat{\mathbf{n}}$. By Newton's third law the satellite gets $-J\hat{\mathbf{n}}$. We apply the impulse–momentum theorem to each body, and its rotational version (torque = lever arm × force) to the satellite. We get

$$m\Delta\mathbf{v}_d = +J\hat{\mathbf{n}}, \qquad M\Delta\mathbf{V} = -J\hat{\mathbf{n}}, \qquad I\Delta\omega = (\mathbf{r}\times(-J\hat{\mathbf{n}}))_z = -Jb, \tag{3.3}$$

where m and M (kg) are the masses of the fragment and the satellite, $\Delta\mathbf{v}_d$ and $\Delta\mathbf{V}$ ($\mathrm{m\,s^{-1}}$) their velocity changes, J (N s) the size of the impulse, I ($\mathrm{kg\,m^2}$) the satellite's moment of inertia about its CoM, $\Delta\omega$ ($\mathrm{rad\,s^{-1}}$) its change in angular velocity, $\mathbf{r}$ (m) the impact point measured from the CoM, and $b = (\mathbf{r}\times\hat{\mathbf{n}})_z$ (m) the lever arm of the impulse about the CoM. The point of the panel that is hit moves with the velocity $\mathbf{V}+\omega\times\mathbf{r}$, so the struck point moves back in two ways: by the motion of the whole satellite and by its rotation. Hence every unit of impulse changes the relative normal speed between the fragment and the struck point by

$$k = \frac{1}{m} + \frac{1}{M} + \frac{b^2}{I}, \tag{3.4}$$

Here k is in $\mathrm{kg^{-1}}$: b^2/I has units $\mathrm{m^2/(kg\,m^2)} = \mathrm{kg^{-1}}$, the same as $1/m$ and $1/M$. The three terms come from the fragment, the satellite moving and the satellite rotating (b appears twice: in the torque and in the speed $v = \omega b$ of the struck point). In an elastic collision the relative normal speed is reversed, as in the one-dimensional case, which gives

$$J = \frac{2v_n}{k}, \tag{3.5}$$

where v_n ($\mathrm{m\,s^{-1}}$) is the normal part of the approach speed. When the impulse passes through the CoM ($b = 0$), $1/k = mM/(m+M)$ and Eqs. (3.3)–(3.5) reduce to the textbook 1-D result for a target initially at rest, $v_1' = \dfrac{m_1-m_2}{m_1+m_2}\,v_1$ and $v_2' = \dfrac{2m_1}{m_1+m_2}\,v_1$ (Ling et al., 2016), where m_1 and m_2 (kg) are the two masses, v_1 ($\mathrm{m\,s^{-1}}$) is the initial speed of body 1 and v_1', v_2' ($\mathrm{m\,s^{-1}}$) are the final speeds. Because the fragment's sideways velocity is unaffected, the fragment rebounds with most of its energy.

3.4. COMPLETELY INELASTIC COLLISION: ONE RIGID BODY

In the completely inelastic case the fragment sticks to the panel, so after the hit the satellite and the fragment are one rigid body. The centre of mass moves a little, to

$$\mathbf{r}_c = \frac{m\mathbf{r}}{m+M}, \tag{3.6}$$

measured from the old CoM. Conservation of linear momentum gives the velocity of the new CoM:

$$m\mathbf{v}_{d0} = (m+M)\,\mathbf{v}_{CM}. \tag{3.7}$$

We find the moment of inertia about the new CoM with the parallel-axis theorem (the old CoM is now $|\mathbf{r}_c|$ away and the fragment is $|\mathbf{r}-\mathbf{r}_c|$ away):

$$I_c = I + M\,|\mathbf{r}_c|^2 + m\,|\mathbf{r}-\mathbf{r}_c|^2. \tag{3.8}$$

Then conservation of angular momentum about the new CoM gives the spin:

$$m\,((\mathbf{r}-\mathbf{r}_c)\times\mathbf{v}_{d0})_z = I_c\,\omega. \tag{3.9}$$

After the hit, every point moves with $\mathbf{v} = \mathbf{v}_{CM} + \omega\times\rho$ (ρ: position from the new CoM). Unlike the elastic case, the sideways momentum is also absorbed, so the satellite gets all of $m\mathbf{v}_{d0}$.

3.5. ENERGY

Kinetic energy is usually not conserved. We start with $K_0 = \frac{1}{2}mv_0^2$, and whatever is not left as kinetic energy of moving or turning is lost:

$$E_{\text{diss}} = K_0 - \left[K_{d1} + \frac{1}{2}M|\Delta\mathbf{V}|^2 + \frac{1}{2}I\omega^2\right], \tag{3.10}$$

where $K_{d1} = \frac{1}{2}m|\mathbf{v}_{d1}|^2$ is the kinetic energy of the fragment after the hit. In the elastic case $E_{\text{diss}} = 0$. In the completely inelastic case it is $K_0 - \frac{1}{2}(m+M)v_{CM}^2 - \frac{1}{2}I_c\omega^2$. For a hit through the CoM this is a fraction $M/(m+M)$ of K_0, the same as a bullet getting stuck in a block (Halliday et al., 2014). When $m \ll M$, almost all of K_0 becomes heat and damage.

4. COMPUTATIONAL AND PYTHON CODING METHODS

4.1. MODEL AND ASSUMPTIONS

Our program calculates what happens to the satellite right after one hit: the impulse it gets, its velocity change $\Delta\mathbf{V}$, its spin ω, the fragment's velocity after the hit, and where the kinetic energy goes. We do not step through time. Instead, the program plugs the inputs into the exact formulas from Section 3. We made these assumptions:

- *The collision is instant.* Outside forces act for too short a time to matter, so $\mathbf{P}$ and L_z are conserved (Eq. (3.2)). We do not model how the contact force changes during the hit.
- *Two dimensions.* Everything moves in the orbital plane and the satellite only spins about z.
- *Rigid satellite, point-mass fragment.* The satellite does not bend or break. The fragment's size is only used to report the diameter of a sphere with the same mass, $d = (6m/\pi\rho)^{1/3}$.
- *Only two ideal types of collision.* Elastic with no friction (Eqs. (3.3)–(3.5)), or completely inelastic (Eqs. (3.6)–(3.9)). A real fragment at these speeds melts or vaporises, so the real answer is somewhere between the two.
- *The orbit is only used for scale.* We only report $|\Delta\mathbf{V}|/v_{\text{orb}}$; we do not calculate the new orbit.

4.2. ALGORITHM

For each hit, the program does these steps.

1. Make the unit vectors $\hat{\mathbf{n}} = (\cos\alpha, \sin\alpha)$ from the panel angle α and $\hat{\mathbf{t}} = \hat{\mathbf{z}} \times \hat{\mathbf{n}}$, then find $\mathbf{v}_{d0}$ from Eq. (3.1).
2. *Elastic:* find $b = r_x n_y - r_y n_x$, k from Eq. (3.4) and J from Eq. (3.5). Then $\Delta\mathbf{V} = -J\hat{\mathbf{n}}/M$, $\omega = -Jb/I$ and $\mathbf{v}_{d1} = \mathbf{v}_{d0} + J\hat{\mathbf{n}}/m$ (Eq. (3.3)).
3. *Completely inelastic:* find $\mathbf{r}_c$, $\mathbf{v}_{CM}$, I_c and ω from Eqs. (3.6)–(3.9). The old CoM of the satellite then moves with $\Delta\mathbf{V} = \mathbf{v}_{CM} + \omega\hat{\mathbf{z}} \times (-\mathbf{r}_c)$, and the stuck fragment moves with $\mathbf{v}_{d1} = \mathbf{v}_{CM} + \omega\hat{\mathbf{z}} \times (\mathbf{r} - \mathbf{r}_c)$.
4. Find K_0, K_{d1}, the satellite's kinetic energy of moving and turning, and E_{diss} (Eq. (3.10)). The impulse on the satellite is reported as $M|\Delta\mathbf{V}|$.
5. Find the speed of a circular orbit. Gravity is the centripetal force, $GM_E M/(R_E + h)^2 = Mv_{\text{orb}}^2/(R_E + h)$, so
$$v_{\text{orb}} = \sqrt{\mu_E/(R_E + h)}, \tag{4.1}$$
where $\mu_E = GM_E$, R_E is Earth's radius and h is the altitude. Then report $|\Delta\mathbf{V}|/v_{\text{orb}}$.
6. Check the conservation laws. The program calculates $\mathbf{P}$ and L_z (Eq. (3.2)) before

and after, and the energy balance, and reports the relative errors

$$\varepsilon_P = \frac{|\mathbf{P}_1 - \mathbf{P}_0|}{|\mathbf{P}_0|}, \quad \varepsilon_L = \frac{|L_{z1} - L_{z0}|}{|\mathbf{P}_0|\,|\mathbf{r}|}, \quad \varepsilon_E = \frac{|K_0 - K_{d1} - K_T - K_R - E_{\text{diss}}|}{K_0}. \quad (4.2)$$

We divide by $|\mathbf{P}_0|\,|\mathbf{r}|$ because L_{z0} can be zero; for a hit at the CoM ($\mathbf{r} = \mathbf{0}$) we set $\varepsilon_L = 0$.

The energy check is a real test. E_{diss} is calculated on its own in the inelastic case and is set to zero in the elastic case, so ε_E is only zero if the impulses in Eqs. (3.3)–(3.9) are correct. For the mass-ratio sweep (head-on, $\mathbf{r} = \mathbf{0}$), the program also compares its answers with the 1-D formulas in Section 3 and with $v = m_1 v_1/(m_1 + m_2)$ for the completely inelastic case.

4.3. PYTHON IMPLEMENTATION

The code is written in Python 3.12 and is run with uv run main.py. It has five files. input_parser.py reads the inputs, changes angles to radians and stops the program if a value makes no physical sense (mass, moment of inertia, speed or altitude not positive, unknown collision type, or impact angle outside $[0^\circ, 90^\circ)$). physics_model.py has the formulas (Eqs. (3.1)–(3.10) and Eq. (4.1)) and the conservation check. solvers.py runs the scenarios, the parameter sweeps and the limiting cases, plotting.py draws the figures, and main.py runs everything and saves the tables. The physics functions work on whole NumPy arrays, so a full sweep is done in one call instead of a loop. We use pandas for the output tables and Matplotlib for the figures. No physical number is typed into the code; all of them come from the input files.

4.4. INPUT AND OUTPUT DESIGN

Each scenario is one JSON file with four parts: *satellite* (M, I, impact point $\mathbf{r}$, panel angle α), *debris* (m, material, v_0, θ), *contact* (elastic or completely_inelastic) and *orbit* (h). Every key name includes its unit (for example mass_kg). A CSV file lists the density of each material, and a JSON file stores μ_E and R_E. Another JSON file sets up the parameter study: a 71×57 grid of debris masses (10^{-6}–10 kg, log spacing) and speeds (1–15 km s^{-1}), 179 impact angles from 0° to 89° for each collision type, 241 mass ratios m/M from 10^{-6} to 100 for each type, and seven limiting cases plus the baseline.

For each scenario, the program saves summary.csv (all results and errors) and conservation.csv (the values of $\mathbf{P}$, L_z and energy before and after). It also saves combined tables for all scenarios and one CSV file per sweep. We designed each figure to show one idea: bar charts of momentum (normal and tangential parts) and energy before and after the hit, a colour map of $|\Delta\mathbf{V}|$ against mass and speed, line graphs of $|\Delta\mathbf{V}|$ and $|\omega|$ against angle and mass ratio, and a bar chart of the conservation errors. We use log scales when a quantity covers many orders of magnitude.

5. DATA AND RESULTS

5.1. INPUT DATA

We set up three hits on the same satellite, using typical values (not measured data) for a small Earth-observation satellite and a millimetre-sized aluminium fragment (Table 5.1). The scenarios differ only in impact angle and collision type: baseline ($\theta = 30^\circ$, completely inelastic), head-on embedding ($\theta = 0^\circ$, completely inelastic) and grazing elastic ($\theta = 80^\circ$, the fragment rebounds). With $\alpha = 90^\circ$, $\hat{\mathbf{n}} = (0,1)$, $\hat{\mathbf{t}} = (-1,0)$ and $b = 4\,\mathrm{m}$. The program also gets the fragment's diameter (8.91 mm), $K_0 = 50\,000\,\mathrm{J}$ and $v_{\mathrm{orb}} = 7585.09\,\mathrm{m\,s^{-1}}$ (Eq. (4.1)).

Quantity	Symbol	Value	Quantity	Symbol	Value
Satellite mass	M	500 kg	Debris density	ρ	$2700\,\mathrm{kg\,m^{-3}}$
Moment of inertia	I	$1200\,\mathrm{kg\,m^2}$	Impact speed	v_0	$10\,000\,\mathrm{m\,s^{-1}}$
Impact point from CoM	$\mathbf{r}$	$(4,0)\,\mathrm{m}$	Orbit altitude	h	550 km
Panel normal from $+x$	α	90°	Earth grav. parameter	μ_E	$3.986 \times 10^{14}\,\mathrm{m^3\,s^{-2}}$
Debris mass (aluminium)	m	0.001 kg	Earth radius	R_E	6 378 137 m

Table 5.1: Input data shared by the three scenarios.

5.2. SCENARIO RESULTS

Table 5.2 shows the results. A negative ω means a clockwise spin about z, which makes sense for a push in the $-y$ direction at $x = +4\,\mathrm{m}$. Velocities are measured in the frame that moved with the satellite before the hit. The conservation errors (Eq. (4.2), last rows) are about 10^{-16}, just round-off: $L_z = -34.641\,\mathrm{kg\,m^2\,s^{-1}}$ both before and after in the baseline, and $|\mathbf{P}| = 10.000\,\mathrm{kg\,m\,s^{-1}}$ in all three scenarios.

Output quantity	Baseline	Head-on	Grazing
Impulse on satellite $M\lvert\Delta\mathbf{V}\rvert$ (N s)	9.99988	9.99985	3.47291
$\lvert\Delta\mathbf{V}\rvert$ ($\mathrm{mm\,s^{-1}}$)	19.9998	19.9997	6.94582
ω ($\mathrm{rad\,s^{-1}}$)	−0.0288671	−0.0333328	−0.0115764
$\lvert\omega\rvert$ ($^\circ\,\mathrm{s^{-1}}$)	1.654	1.910	0.663
Fragment velocity after, $\mathbf{v}_{d1}$ ($\mathrm{m\,s^{-1}}$)	(0.0100, −0.133)	(0, −0.153)	(9848.08, 1736.43)
Fragment K_{d1} (J)	8.9×10^{-6}	1.2×10^{-5}	49 999.908
Satellite K_T (J)	0.100	0.100	0.0121
Satellite K_R (J)	0.500	0.667	0.0804
E_{diss} (J)	49 999.400	49 999.233	0
E_{diss}/K_0	0.999988	0.999985	0
$\lvert\Delta\mathbf{V}\rvert/v_{\mathrm{orb}}$	2.64×10^{-6}	2.64×10^{-6}	9.16×10^{-7}
Residual ε_P (Eq. (4.2))	8.9×10^{-17}	0	0
Residual ε_L	1.8×10^{-16}	3.6×10^{-16}	0
Residual ε_E	0	0	1.5×10^{-16}

Table 5.2: Calculated results for the three scenarios.

Sample calculation (baseline, completely inelastic).

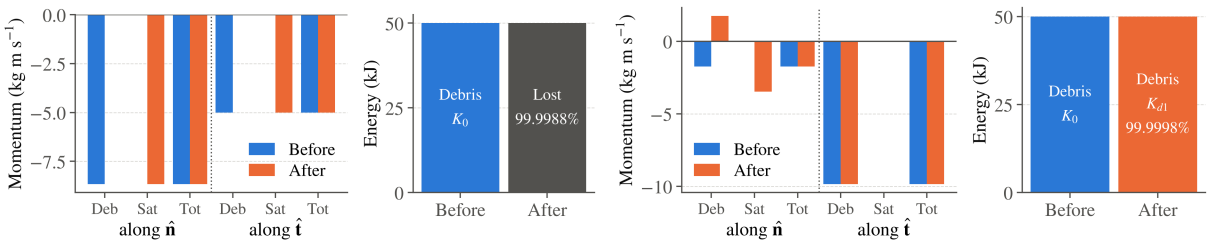
From Eq. (3.1), $\mathbf{v}_{d0} = (5000,\,-8660.25)\,\mathrm{m\,s^{-1}}$. Equations (3.6)–(3.9) give

$$\begin{aligned}
\mathbf{r}_c &= \frac{(0.001)(4,0)}{500.001} = (8.0\times10^{-6},\,0)\,\mathrm{m},\\
\mathbf{v}_{CM} &= \frac{0.001\,\mathbf{v}_{d0}}{500.001} = (0.0100,\,-0.01732)\,\mathrm{m\,s^{-1}},\\
I_c &= 1200 + 500(8.0\times10^{-6})^2 + 0.001(3.999992)^2 = 1200.016\,\mathrm{kg\,m^2},\\
\omega &= \frac{0.001\,(3.999992)(-8660.25)}{1200.016} = -0.028867\,\mathrm{rad\,s^{-1}},\\
E_{\mathrm{diss}} &= 50\,000 - \tfrac{1}{2}(500.001)(0.02000)^2 - \tfrac{1}{2}(1200.016)(0.028867)^2\\
&= 49\,999.40\,\mathrm{J}.
\end{aligned} \tag{5.1}$$

Sample calculation (grazing, elastic).

$v_n = 10\,000\cos 80^\circ = 1736.48\,\mathrm{m\,s^{-1}}$, $k = 1000 + 0.002 + 16/1200 = 1000.0153\,\mathrm{kg^{-1}}$ (Eq. (3.4)), so $J = 2(1736.48)/1000.0153 = 3.47291\,\mathrm{N\,s}$, $|\Delta\mathbf{V}| = J/M = 6.94582\,\mathrm{mm\,s^{-1}}$ and $\omega = -Jb/I = -0.0115764\,\mathrm{rad\,s^{-1}}$ (Eqs. (3.3)–(3.5)).

Figure 5.1 shows where the momentum and energy go. In the baseline the satellite gets both the normal and sideways momentum and almost all of K_0 is lost; in the grazing elastic case only the normal momentum is reversed and the fragment keeps almost all its energy.



(a) Baseline (30°, completely inelastic). **(b)** Grazing (80°, elastic).

Figure 5.1: Momentum along $\hat{\mathbf{n}}$ and $\hat{\mathbf{t}}$ (left of each plot; Deb = debris, Sat = satellite, Tot = total) and energy budget (right).

5.3. PARAMETRIC STUDY

Figure 5.2 shows $|\Delta\mathbf{V}|$ for all 4047 mass–speed pairs with the baseline geometry. It goes from $2.0\times10^{-6}\,\mathrm{m\,s^{-1}}$ (1 mg at $1\,\mathrm{km\,s^{-1}}$) up to $269\,\mathrm{m\,s^{-1}}$ (10 kg at $15\,\mathrm{km\,s^{-1}}$). When $m \ll M$, $|\Delta\mathbf{V}| \approx mv_0/M$, so it grows in direct proportion to both mass and speed.

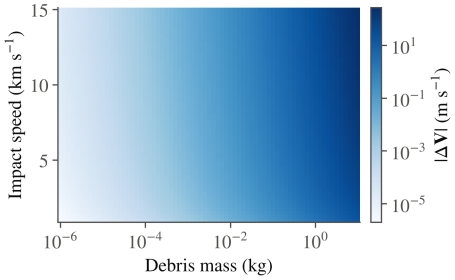


Figure 5.2: $|\Delta\mathbf{V}|$ versus debris mass and speed (baseline, log scales).

Case	$\lvert\Delta\mathbf{V}\rvert$ (m s^{-1})	ω (rad s^{-1})	E_{diss}/K_0
1 µg fragment	2.0×10^{-8}	-2.9×10^{-8}	1.000000
$m=M$, head-on, elastic	10 000	0	0
$m=M$, head-on, inelastic	5000	0	0.5
$m=100M$, head-on, elastic	19 801.98	0	0
Grazing, 89°	0.0200	-5.8×10^{-4}	0.999998
Hit at CoM ($\mathbf{r}=\mathbf{0}$)	0.0200	0	0.999998
Baseline, elastic	0.0346	−0.0577	0

Table 5.3: Limiting cases (baseline values unless stated).

Figure 5.3 shows how the impact angle matters. In the completely inelastic case $|\Delta\mathbf{V}| = 20.0\,\text{mm}\,\text{s}^{-1}$ at every angle, because the satellite absorbs the whole momentum $m\mathbf{v}_{d0}$. In the elastic case only the normal momentum is reversed, so $|\Delta\mathbf{V}| \approx 2mv_0\cos\theta/M$ drops from 40.0 to $0.70\,\text{mm}\,\text{s}^{-1}$. The two curves cross at $\theta = 60°$, where $2\cos\theta = 1$. The sideways velocity points along $\mathbf{r}$ and gives no torque, so the spin depends only on the normal part: $|\omega| \propto \cos\theta$ in both cases, the elastic spin about twice the inelastic one.

Figure 5.4 gives the familiar 1-D collision. When $m \ll M$, the satellite's velocity grows in proportion to m/M (twice as fast in the elastic case), and in the elastic case the fragment bounces back at $-v_0$. When $m \gg M$, the target reaches $2v_0$ (elastic) or v_0 (completely inelastic). Over all 482 points, the biggest difference from the 1-D formulas is $6.7\times10^{-16}\,v_0$. Table 5.3 lists the limiting cases.

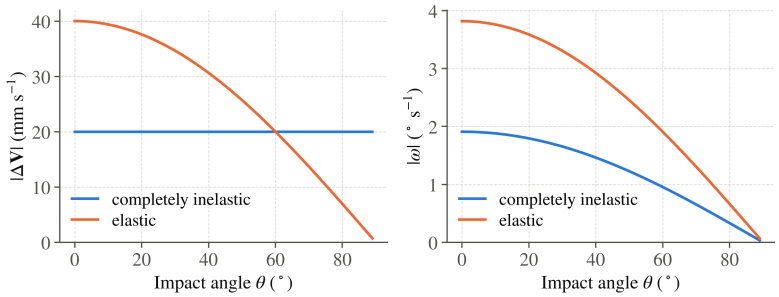


Figure 5.3: Satellite velocity change (left) and spin rate (right) versus impact angle for both collision types (179 angles each).

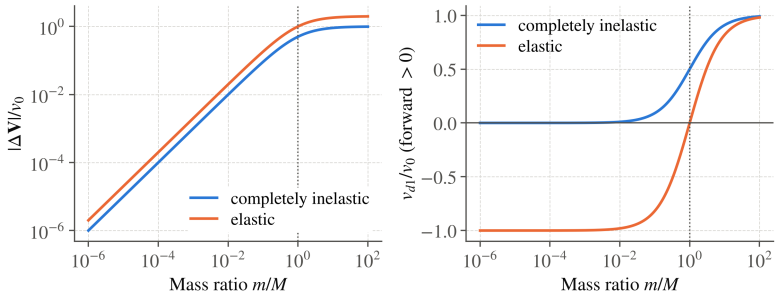


Figure 5.4: Head-on hit through the CoM: satellite velocity (left) and fragment velocity after the impact (right), both divided by v_0, versus the mass ratio m/M (241 ratios per collision type).

6. DISCUSSION AND CONCLUSION

6.1. DISCUSSION

Comparing with the hypothesis.

Our results agree with all three parts of the hypothesis (Section 2). In the baseline, a 1 g fragment changes the satellite's velocity by $20.0\,\mathrm{mm\,s^{-1}}$, which is only 2.6×10^{-6} of the orbital speed (Table 5.2). So one hit like this does not really change the orbit. The spin is $1.65^\circ\,\mathrm{s^{-1}}$, about what we expected. The angular momentum $mv_0\cos\theta\, b = 34.6\,\mathrm{kg\,m^2\,s^{-1}}$ is large because the hit is at the tip of the panel, 4 m from the CoM. At this rate the satellite would turn a full circle in less than four minutes if nothing stopped it. Finally, 99.9988% of the 50 kJ is lost. This is a little less than the $M/(m+M) = 99.9998\%$ for a hit through the CoM, because 0.5 J goes into rotation. The satellite gets less than 1 J of kinetic energy in total. In other words, the momentum is passed on very well, but the energy is not.

What the sweeps show.

The angle sweep (Fig. 5.3) shows that the type of collision changes not only how much momentum is passed on, but also which part. A fragment that sticks always gives all of its momentum, so $|\Delta\mathbf{V}|$ does not depend on θ. A fragment that bounces off only doubles the normal part, so it gives a bigger push than a sticking fragment only when $\theta < 60^\circ$. The spin depends only on the normal impulse in our geometry, so it drops like $\cos\theta$ in both cases. The mass–speed map (Fig. 5.2) grows linearly with m and v_0 as long as $m \ll M$. Its top corner (kilograms at $15\,\mathrm{km\,s^{-1}}$, up to 1.1 GJ) is outside what our model can describe: a hit like that would break the satellite apart instead of just pushing it.

How precise are the inputs?

When $m \ll M$, $|\Delta\mathbf{V}|$ and ω are proportional to m and v_0. So a 10% error in the input mass or speed gives a 10% error in the results, and E_{diss} doubles any relative error in v_0. The computer's precision is not a problem: the errors of 10^{-16} (Table 5.2) and the match with the 1-D formulas to 6.7×10^{-16} show that the code solves the equations exactly. What limits the accuracy is the model itself. The satellite is rigid and 2-D, the fragment is a point, and a real hit at these speeds is somewhere between elastic and completely inelastic. Material thrown back out of the crater can even make the momentum transfer bigger than in the completely inelastic case, and our model does not include this.

Comparing with a real event.

For the Sentinel-1A hit, Krag et al. (2017) used GPS and attitude data to estimate a velocity change of only about $0.7\,\mathrm{mm\,s^{-1}}$ and a peak rotation rate of about $0.04^\circ\,\mathrm{s^{-1}}$ about the yaw axis. Our numbers are bigger because our satellite is lighter, with a smaller moment of inertia, and because we chose the fragment and its speed ourselves. But the overall picture is the same: the velocity change was tiny compared with the

orbital speed, while the change in orientation could clearly be measured. This supports our conclusion that, for small debris, the spin from an off-centre hit matters more than the change in orbit.

6.2. APPLICATIONS

The same momentum and energy calculations are useful in real engineering.

- *Attitude control.* The spin from Eq. (3.9) is a disturbance that the reaction wheels or thrusters have to cancel. The angular momentum from a hit, $mv_n b$, tells engineers how much the wheels must be able to absorb. For example, Sentinel-1A had a sudden change of orientation after its 2016 hit (Krag et al., 2017).
- *Keeping the orbit.* Comparing $|\Delta\mathbf{V}|$ with v_{orb} tells us whether a hit needs an orbit correction. For debris from millimetres to centimetres in size the change is tiny, but a fragment of several kilograms would need a correction burn.
- *Shielding and panel design.* Equation (3.10) shows that in an inelastic hit almost all the energy is released at one spot. This is why spacecraft use spaced shields (Whipple shields) that break the fragment apart before it reaches the main wall, and why solar panels are built so that one crater only knocks out part of the power.
- *The debris problem.* Every hit can create new fragments. Working out the energy and momentum of collisions is the first step in models of the debris population and of the chain reaction predicted by Kessler & Cour-Palais (1978).

6.3. CONCLUSION

Using conservation of linear and angular momentum together with the impulse–momentum theorem and the parallel-axis theorem, we could solve exactly what happens when a debris fragment hits a solar panel, for both the elastic and the completely inelastic case. Our Python program shows that a 1 g fragment at $10\,\mathrm{km\,s^{-1}}$ changes the velocity of a 500 kg satellite by only $20\,\mathrm{mm\,s^{-1}}$ (2.6×10^{-6} of its orbital speed). However, it makes the satellite spin at $1.65^\circ\,\mathrm{s^{-1}}$, and almost all of the 50 kJ impact energy is lost at the panel. These results confirm our hypothesis, show the same pattern as the Sentinel-1A measurements, and pass the conservation checks to round-off level. As a next step, the two ideal cases could be replaced by a measured momentum-transfer value, and the orbit after the hit could be calculated.

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